

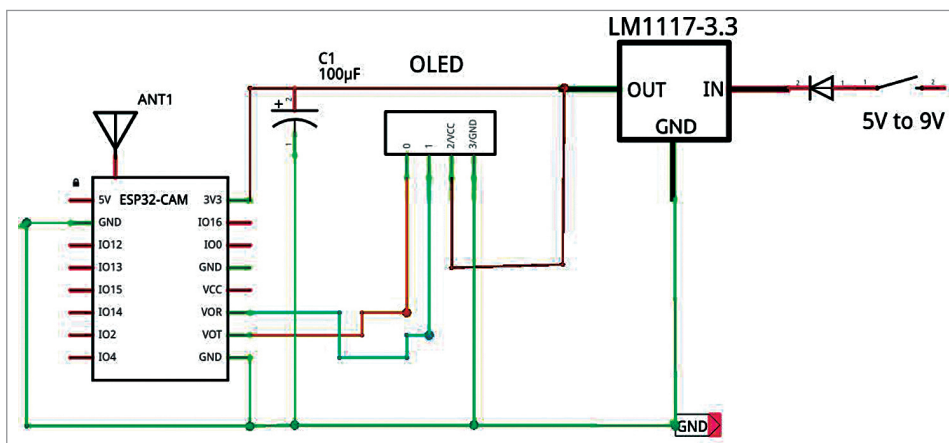


Fig. 3: Circuit diagram of ESP32 camera webserver

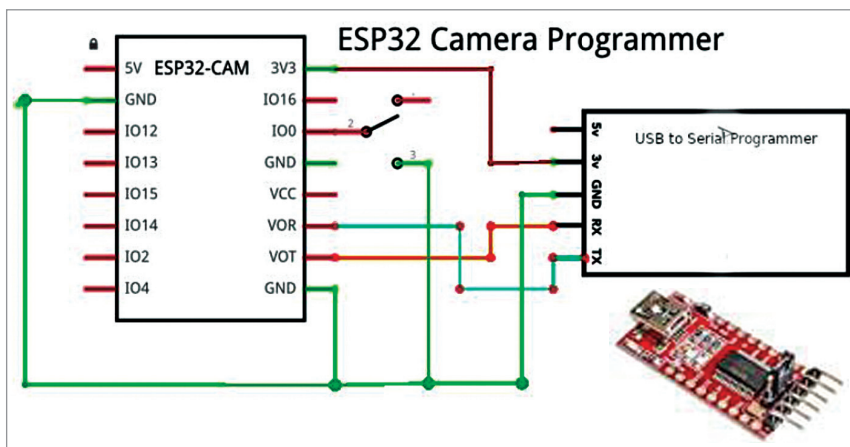


Fig. 4: Schematic of ESP32 cam programming

Circuit and working

The circuit diagram for ESP32 camera webserver is shown in Fig. 3. The circuit is built around voltage regulator LM1117, an OLED, and



Fig. 5: Webcam with OLED to display

ESP32 cam. For programming ESP32 you need the USB-to-serial programmer shown in Fig. 4.

Software

The ESP32 camera does not come with built-in USB-to-serial interface. Therefore, to upload the sketch you need a USB-to-serial programmer. The schematic for the programmer is shown in Fig. 3.

The GPIO 0 pin is to be pulled to ground during uploading mode and reset button of the ESP32 needs to be pressed for first few seconds of uploading. After uploading the

source code esp32-webcam-server.ino into ESP32 board, remove the board and use it in the circuit.

The device can be powered using any regulated 5V DC adaptor, or a 6V battery, or a voltage regulator circuit. The SD card module on ESP board uses GPIO 4, GPIO 12, and GPIO 13. Only GPIO 1 and GPIO 3 are the free pins available, which can be used as serial pins for uploading the program in flash mode.

The GPIO1 and GPIO 3 pins are free after uploading the program, so we can use them as I2C pins for connecting the OLED display.

Construction and testing

Upload the source code esp32-webcam-server.ino into ESP32 as described above. Assemble the circuit as per Fig. 2. On powering the device to connect it to the Wi-Fi available in the network, it will show the IP address on the OLED display connected to it.

Now open any web browser in your smartphone or laptop and search the IP address displayed on OLED display. You should be able to see the live video on your phone/laptop. The webcam with OLED to display its network name and IP address is shown in Fig. 5.

The low-cost ESP32 camera has wide possibilities because it is programmable. By fixing a series of these cameras in line, we can get a wide object scanner of 50 to 100 metres for taking pictures of approaching vehicles' number plates to check for over speeding. In solo mode, with a PIR sensor, it can be used as a bird snap catcher. For safety against Covid, it can be used as a scanner for high temperature detection. **EFY**

EFY Note

The source code of this project is available for free download at www.electronicsforu.com



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